

Optical Interference:

Comparison of Gaussian and Laguerre-Gaussian Beams

A project dissertation submitted in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE

in

PHYSICS

by

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Declaration

I hereby declare that the project dissertation entitled “**Optical Interference: Comparison of Gaussian and Laguerre-Gaussian beams**” submitted by me to the **Department of Physics, Aliah University**, Newtown, Kolkata - 700160 in partial fulfillment of the requirements for the award of **Master of Science in Physics** is a bona-fide record of the work carried out by me under the supervision of **Dr. Md. Mabud Hossain**.

I further declare that the work reported in this dissertation, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

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Certificate

This is to certify that the project dissertation entitled “Optical Interference: Comparison of Gaussian and Laguerre-Gaussian beams” submitted by PRIYAM GHOSH (Roll No. PHY212058) to Aliah University, in partial fulfillment of the requirement for the award of the degree of Master of Science in Physics is a bona-fide work carried out under my supervision. The dissertation fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission.

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CONTENTS

DECLARATION

CERTIFICATE

ACKNOWLEDGEMENT

1. INTRODUCTION

- 1.1 PROPERTIES OF LIGHT: PARTICLE NATURE OF LIGHT
- 1.2 WAVE NATURE OF LIGHT
- 1.3 ELECTROMAGNETIC THEORY OF LIGHT
- 1.4 QUANTUM THEORY OF LIGHT
- 1.5 MODERN THEORY OF LIGHT
- 1.6 GAUSSIAN NATURE OF LIGHT
- 1.7 LAGUERRE – GAUSSIAN
- 1.8 INTERFERENCE OF LIGHT
- 1.9 LAGUERRE – GAUSSIAN (LG) BEAM: ORIGIN & PROPERTIES

2. INTERFERENCE BETWEEN TWO WAVES

- 2.1 GAUSSIAN WAVE
- 2.2 LAGUERRE – GAUSSIAN BEAM

3. GRAPHICAL REPRESENTATION OF DIFFERENT INTERFERENCE PATTERN FOR GAUSSIAN & LAGUERRE – GAUSSIAN BEAM

4. CONCLUSION

5. REFERENCE

INTRODUCTION

Light is an electromagnetic radiation that can be detected by the human eye. Electromagnetic radiation occurs over an extremely wide range of wavelengths, from gamma rays with wavelengths less than about 1×10^{-11} meter to radio waves measured in meters. Within that broad spectrum the wavelengths visible to humans occupy a very narrow band, from about 700 nanometers (nm; billionths of a meter) for red light down to about 400 nm for violet light. The spectral regions adjacent to the visible band are often referred to as light also, infrared at the one end and ultraviolet at the other. The speed of light in a vacuum is a fundamental physical constant, the currently accepted value of which is exactly 299,792,458 meters per second, or about 186,282 miles per second.



<https://www.google.com/url?sa=i&url=https%3A%2F%2Fen.wikipedia.org%2Fwiki%2FLight&psig=AOvVaw3nG6Tk>

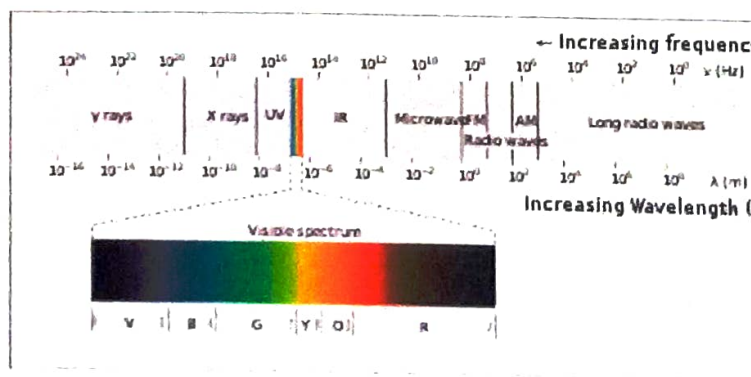
PROPERTIES OF LIGHT: PARTICLE NATURE

Pierre Gassendi (1592–1655), proposed a particle theory of light which was published posthumously in the 1660s. Isaac Newton studied Gassendi's work at an early age and preferred his view to Descartes's theory of the plenum. He stated in his Hypothesis of Light of 1675 that

light was composed of corpuscles (particles of matter) which were emitted in all directions from a source. One of Newton's arguments against the wave nature of light was that waves were known to bend around obstacles, while light travelled only in straight lines. He did, however, explain the phenomenon of the diffraction of light (which had been observed by Francesco Grimaldi) by allowing that a light particle could create a localized wave in the ether.

Newton's theory could be used to predict the reflection of light, but could only explain refraction by incorrectly assuming that light accelerated upon entering a denser medium because the gravitational pull was greater. Newton published the final version of his theory in his *Opticks* of 1704. His reputation helped the particle theory of light to hold sway during the eighteenth century. The particle theory of light led Laplace to argue that a body could be so massive that light could not escape from it. In other words, it would become what is now called a black hole. Laplace withdrew his suggestion later, after a wave theory of light became firmly established as the model for light (as has been explained, neither a particle or wave theory is fully correct). A translation of Newton's essay on light appears in the large scale structure of space-time, by Stephen Hawking and George F. R. Ellis.

The fact that light could be polarized was for the first time qualitatively explained by Newton using the particle theory. Étienne-Louis Malus in 1810 created a mathematical particle theory of polarization. Jean-Baptiste Biot in 1812 showed that this theory explained all known phenomena of light polarization. At that time the polarization was considered as the proof of the particle theory.

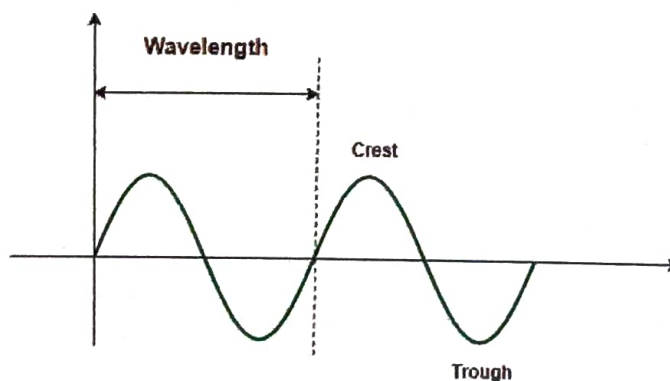


https://en.wikipedia.org/wiki/File:EM_spectrum.svg

WAVE NATURE OF LIGHT

To explain the origin of colors, Robert Hooke (1635–1703) developed a "pulse theory" and compared the spreading of light to that of waves in water in his 1665 work *Micrographia* ("Observation IX"). In 1672 Hooke suggested that light's vibrations could be perpendicular to the direction of propagation. Christiaan Huygens (1629–1695) worked out a mathematical wave theory of light in 1678 and published it in his *Treatise on Light* in 1690. He proposed that light was emitted in all directions as a series of waves in a medium called the luminiferous aether. As waves are not affected by gravity, it was assumed that they slowed down upon entering a denser medium. The wave theory predicted that light waves could interfere with each other like sound waves (as noted around 1800 by Thomas Young). Young showed by means of a diffraction experiment that light behaved as waves. He also proposed that different colors were caused by different wavelengths of light and explained color vision in terms of three-colored receptors in the eye. Another supporter of the wave theory was Leonhard Euler. He argued in *Nova theoria lucis et colorum* (1746) that diffraction could more easily be explained by a wave theory.

The weakness of the wave theory was that light waves, like sound waves, would need a medium for transmission. The existence of the hypothetical substance luminiferous aether proposed by Huygens in 1678 was cast into strong doubt in the late nineteenth century by the Michelson–Morley experiment.

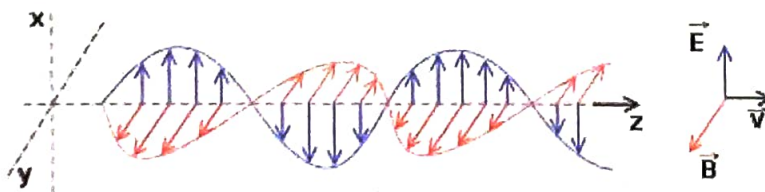


GeeksforGeek <https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.geeksforgeeks.org%2Flight-waves->

ELECTROMAGNETIC THEORY OF LIGHT

In 1845, Michael Faraday discovered that the plane of polarization of linearly polarized light is rotated when the light rays travel along the magnetic field direction in the presence of a transparent dielectric, an effect now known as Faraday rotation. This was the first evidence that light was related to electromagnetism.

Faraday's work inspired James Clerk Maxwell to study electromagnetic radiation and light. Maxwell discovered that self-propagating electromagnetic waves would travel through space at a constant speed, which happened to be equal to the previously measured speed of light. From this, Maxwell concluded that light was a form of electromagnetic radiation: he first stated this result in 1862 in *On Physical Lines of Force*. In 1873, he published *A Treatise on Electricity and Magnetism*, which contained a full mathematical description of the behavior of electric and magnetic fields, still known as Maxwell's equations. Soon after, Heinrich Hertz confirmed Maxwell's theory experimentally by generating and detecting radio waves in the laboratory and demonstrating that these waves behaved exactly like visible light, exhibiting properties such as reflection, refraction, diffraction and interference.



https://en.wikipedia.org/wiki/File:Onde_electromagnetique.svg

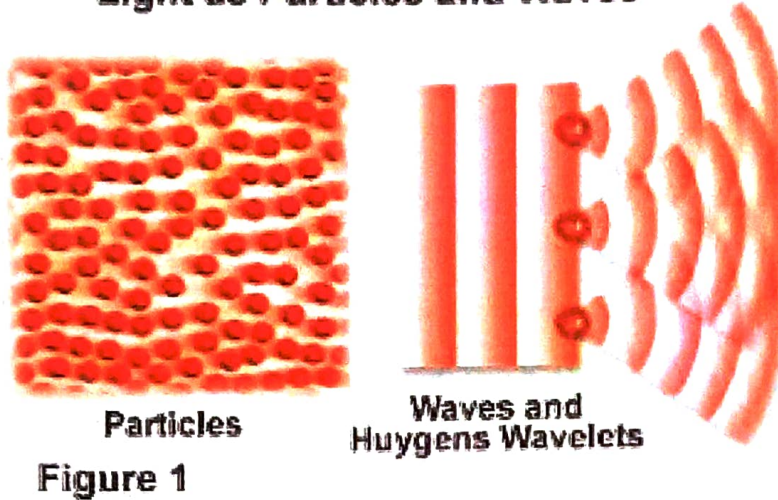
QUANTUM THEORY OF LIGHT

In 1900 Max Planck, attempting to explain black-body radiation, suggested that although light was a wave, these waves could gain or lose energy only in finite amounts related to their frequency. Planck called these "lumps" of light energy "quanta" (from a Latin word for "how much"). In 1905, Albert Einstein used the idea of light quanta to explain the photoelectric effect and suggested that these light quanta had a "real" existence. In 1923 Arthur Holly Compton showed that the wavelength shift seen when low intensity X-rays scattered from electrons (so called Compton scattering) could be explained by a particle-theory of X-rays, but not a wave theory. In 1926 Gilbert N. Lewis named these light quanta particles photons.

MODERN THEORY ON LIGHT

In most everyday circumstances, the properties of light can be derived from the theory of classical electromagnetism, in which light is described as coupled electric and magnetic fields propagating through space as a traveling wave. However, this wave theory, developed in the mid-19th century, is not sufficient to explain the properties of light at very low intensities. At that level a quantum theory is needed to explain the characteristics of light and to explain the interactions of light with atoms and molecules. In its simplest form, quantum theory describes light as consisting of discrete packets of energy, called photons. However, neither a classical wave model nor a classical particle model correctly describes light; light has a dual nature that is revealed only in quantum mechanics. This surprising wave-particle duality is shared by all of the primary constituents of nature (e.g., electrons have both particle-like and wavelike aspects). Since the mid-20th century, a more comprehensive theory of light, known as quantum electrodynamics (QED), has been regarded by physicists as complete. QED combines the ideas of classical electromagnetism, quantum mechanics, and the special theory of relativity.

Light as Particles and Waves



<https://static5.olympus-lifescience.com/data/olympusmicro/primer/lightandcolor/images/particlewavefigure1.jpg?rev=1630>

GAUSSIAN NATURE OF LIGHT

In optics, a Gaussian beam is a beam of electromagnetic radiation with high monochromaticity whose amplitude envelope in the transverse plane is given by a Gaussian function; this also implies a Gaussian intensity (irradiance) profile. This fundamental (or TEM₀₀) transverse Gaussian mode describes the intended output of most (but not all) lasers, as such a beam can be focused into the most concentrated spot. When such a beam is refocused by a lens, the transverse phase dependence is altered; this results in a different Gaussian beam. The electric and magnetic field amplitude profiles along any such circular Gaussian beam (for a given wavelength and polarization) are determined by a single parameter: the so-called waist w_0 .

The Gaussian beam is a transverse electromagnetic (TEM) mode. The mathematical expression for the electric field amplitude is a solution to the paraxial Helmholtz equation. Assuming polarization in the x direction and propagation in the +z direction, the electric field in phasor (complex) notation is given by:

$$\mathbf{E}(r, z) = E_0 \hat{\mathbf{x}} \frac{w_0}{w(z)} \exp\left(\frac{-r^2}{w(z)^2}\right) \exp\left(-i \left(kz + k \frac{r^2}{2R(z)} - \psi(z) \right)\right)$$

r is the radial distance from the center axis of the beam,

z is the axial distance from the beam's focus (or "waist"),

i is the imaginary unit,

$k = 2\pi n/\lambda$ is the wave number (in radians per meter) for a free-space wavelength λ , and n is the index of refraction of the medium in which the beam propagates,

$E_0 = E(0, 0)$, the electric field amplitude (and phase) at the origin ($r = 0, z = 0$),

$w(z)$ is the radius at which the field amplitudes fall to $1/e$ of their axial values (i.e., where the intensity values fall to $1/e^2$ of their axial values), at the plane z along the beam,

$w_0 = w(0)$ is the waist radius,

$R(z)$ is the radius of curvature of the beam's wavefronts at z , and

$\psi(z)$ is the Gouy phase at z , an extra phase term beyond that attributable to the phase velocity of light.

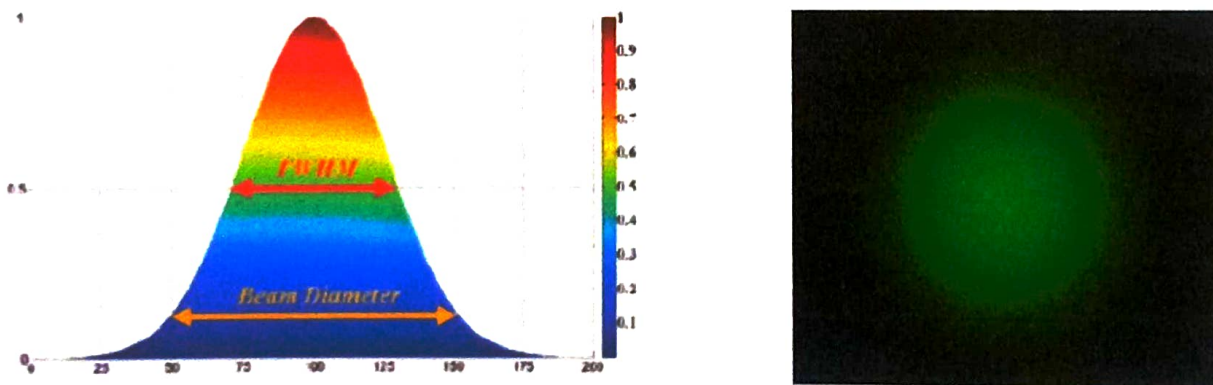


Figure: Intensity profile of the Gaussian beam

https://en.wikipedia.org/wiki/File:Green_laser_pointer_TEM00_profile.JPG

Laguerre-Gaussian Mode

Beam profiles which are circularly symmetric (or lasers with cavities that are cylindrically symmetric) are often best solved using the Laguerre-Gaussian modal decomposition. These functions are written in cylindrical coordinates using generalized Laguerre polynomials. Each transverse mode is again labeled using two integers, in this case the radial index $p \geq 0$ and the azimuthal index l which can be positive or negative (or zero):

$$u(r, \phi, z) = C_{lp}^{LG} \frac{1}{w(z)} \left(\frac{r\sqrt{2}}{w(z)} \right)^{|l|} \exp\left(-\frac{r^2}{w^2(z)}\right) L_p^{|l|} \left(\frac{2r^2}{w^2(z)} \right) \times \\ \exp\left(-ik\frac{r^2}{2R(z)}\right) \exp(-il\phi) \exp(i\psi(z)),$$

Where L_p^l is the generalized Laguerre polynomials and C_{lp}^{LG} is a required normalization constant:

$$C_{lp}^{LG} = \sqrt{\frac{2p!}{\pi(p+|l|)!}} \Rightarrow \int_0^{2\pi} d\phi \int_0^\infty r dr |u(r, \phi, z)|^2 = 1$$

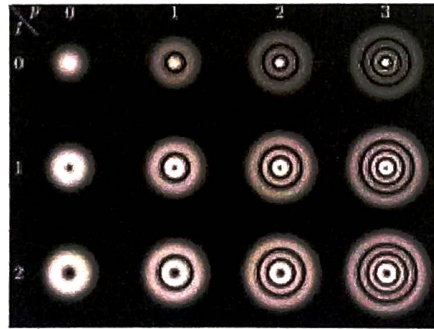


Figure: Intensity profiles of the Laguerre-Gaussian modes.

https://en.wikipedia.org/wiki/File:Intensity_profiles_of_Laguerre-Gaussian_modes.pdf

INTERFERENCE OF LIGHT

In physics, interference is a phenomenon in which two coherent waves are combined by adding their intensities or displacements with due consideration for their phase difference. The resultant wave may have greater intensity (constructive interference) or lower amplitude (destructive interference) if the two waves are in phase or out of phase, respectively. Interference effects can be observed with all types of waves, for example, light, radio, acoustic, surface water waves, gravity waves, or matter waves as well as in loudspeakers as electrical waves.

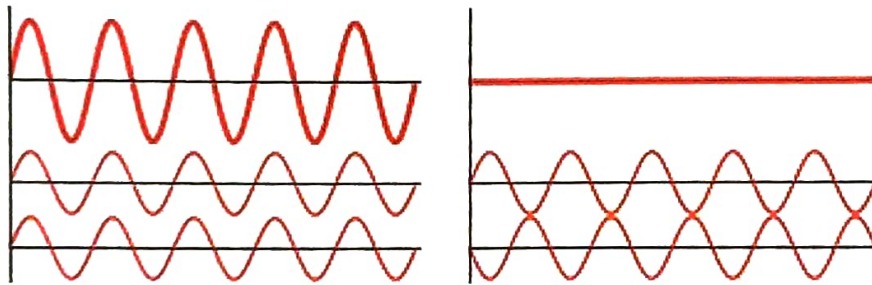


Figure: The interference of two waves. When in phase, the two lower waves create constructive interference (left), resulting in a wave of added amplitude. When 180° out of phase, they create destructive interference (right), resulting in a wave of zero amplitude.
(https://en.wikipedia.org/wiki/File:Interference_of_two_waves.svg)

LAGUERRE – GAUSSIAN (LG) BEAM: ORIGIN & PROPERTIES

We know that in general light travels with plane wavefront. However if the light have a twisted wavefront instead where Poynting vector is not parallel with the wavefront , the resulting form of light can be created by adjusting different parameter. This type of light where one can control the pattern of light by controlling different parameters is called Structured light. By solving paraxial Helmholtz equation in cylindrical coordinate the resulting solution is called Laguerre-Gaussian

beam, which is a type of structured light. We also know that a beam of light possess both orbital angular momentum (OAM) and spin angular momentum (SAM) because of spatial distribution of wavefront and polarization respectively. A Laguerre-Gaussian beam carries an orbital angular momentum proportional to the phase of the light beam. In this article we will theoretically calculate the interference of a Laguerre-Gaussian beam and compare it with normal Gaussian beam which actually is a Laguerre-Gaussian beam with no orbital angular momentum.

The interference of two Laguerre-Gaussian (LG) beams is an important topic in the field of optics, as it has numerous practical applications, ranging from laser physics and optical communication to microscopy and material processing. LG beams are a type of laser beam that have a characteristic doughnut-shaped intensity profile and carry orbital angular momentum (OAM). When two LG beams with the same OAM intersect, they interfere and create a complex interference pattern that depends on several parameters, such as the relative phase, the beam waist, and the radial and azimuthal mode indices. The study of LG beam interference has attracted considerable attention in recent years, both from a fundamental and an applied perspective, and has led to the development of new tools and techniques for manipulating light at the nanoscale level.

INTERFERENCE BETWEEN TWO WAVES

GAUSSIAN WAVE

Equation of the Gaussian wave can be written as ,

$$Y=Ae^{i(\omega t-kx)}$$

Now let us consider two Gaussian waves forming an interference pattern,

$$Y_1=A_1e^{i(\omega t-kx_1)} \dots\dots (1)$$

$$Y_2=A_2e^{i(\omega t-kx_2)} \dots\dots (2)$$

According to the principle of superposition,

$$\begin{aligned}
 Y &= Y_1 + Y_2 \\
 &= e^{i\omega t} (A_1 e^{-ikx_1} + A_2 e^{-ikx_2}) \\
 &= e^{i\omega t} \{A_1 (\cos kx_1 - i \sin kx_1) + A_2 (\cos kx_2 - i \sin kx_2)\} \\
 &= e^{i\omega t} (a - ib) = e^{i\omega t} \sqrt{a^2 + b^2} \cdot e^{-i\Phi}
 \end{aligned}$$

where $\Phi = \tan^{-1}(b/a)$, $a = A_1 \cos kx_1 + A_2 \cos kx_2$, $b = A_1 \sin kx_1 + A_2 \sin kx_2$

Therefore, $Y = Ae^{i(\omega t - \Phi)}$, where $A^2 = a^2 + b^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos k(x_2 - x_1)$ (3)

Here A is the amplitude of the resultant disturbance. Hence the intensity of light,

$$\begin{aligned}
 I &\propto A^2 \\
 &= a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta \quad \text{.....(4)}
 \end{aligned}$$

Where $\delta = k.(x_1 - x_2) = (2\pi/\lambda)(x_1 - x_2)$ is the phase difference between 2 interfering waves.

LAGUERRE - GAUSSIAN BEAM

Equation of LG wave can be written as ,

$$E(r, \Phi, z=0) = E_0 (\sqrt{2r/\omega_0})^{|l|} e^{il\Phi} e^{-\{(r^2)/\omega_0\}} \quad \text{.....(5)}$$

Let us consider two LG waves ,

$$E_1 = E_0 (\sqrt{2r/\omega_0})^{|l_1|} e^{il_1 \Phi_1} e^{-\{(r^2)/\omega_0\}} e^{i(kz + \theta_1)} \quad \text{.....(6)}$$

$$E_2 = E_0 (\sqrt{2r/\omega_0})^{|l_2|} e^{il_2 \Phi_2} e^{-\{(r^2)/\omega_0\}} e^{i(kz + \theta_2)} \quad \text{.....(7)}$$

As seen with Gaussian wave interference we can write ,

$$E=E_1+E_2$$

$$E=E_1^*+E_2^*+2E_1^*E_2^*\cos(\delta_1-\delta_2) \dots\dots(8)$$

Where

$$E_1^*=E_0(\sqrt{2r/\omega_0})_1^l e^{-\{(r^2)/\omega_0\}}$$

$$E_2^*=E_0(\sqrt{2r/\omega_0})_2^l e^{-\{(r^2)/\omega_0\}}$$

$$\delta_1=l_1 \Phi_1-\theta_1$$

$$\delta_2=l_2 \Phi_2-\theta_2$$

Now $\delta_1-\delta_2$ can be written as , $\delta_{12} = (\delta_1-\delta_2)$

$$=l_1 \Phi_1-\theta_1-l_2 \Phi_2+\theta_2$$

$$=(l_1 \Phi_1-l_2 \Phi_2)+(\theta_2-\theta_1) \dots\dots(9)$$

here δ_{12} is the phase difference between the 2 LG wave

Phase between two LG beam can also be written as ,

$$\delta_{12}=k.(x_2-x_1)$$

where x_2-x_1 = path difference and k =wave number

$$\delta_{12}=(2\pi/\lambda).(x_2-x_1)$$

Now previously we have seen,

$$I \propto A^2$$

Therefore it can be written as,

$$I=E^2 + E^2 + 2E^2\cos\delta_{12}$$

(after considering $E_1^*=E_2^*=E$)

$$=2E^2+2E^2\cos\delta_{12}$$

$$=2E^2(1+\cos\delta_{12})$$

$$=2E^22\cos^2(\delta_{12}/2)$$

$$=4E^2\cos^2(\delta_{12}/2)$$

Therefore, we have found

$$I \propto \cos^2(\delta_{12}/2) \dots\dots(10)$$

Equation (10) can be alternatively written as ,

$$I \propto \cos^2\{(\pi ax)/(\lambda d)\} \dots\dots(11)$$

a is width of slit , d is the distance between slit and screen.

GRAPHICAL REPRESENTATION OF DIFFERENT INTERFERENCE PATTERN FOR GAUSSIAN & LAGUERRE – GAUSSIAN BEAMS

TWO DIMENSIONAL PLOT

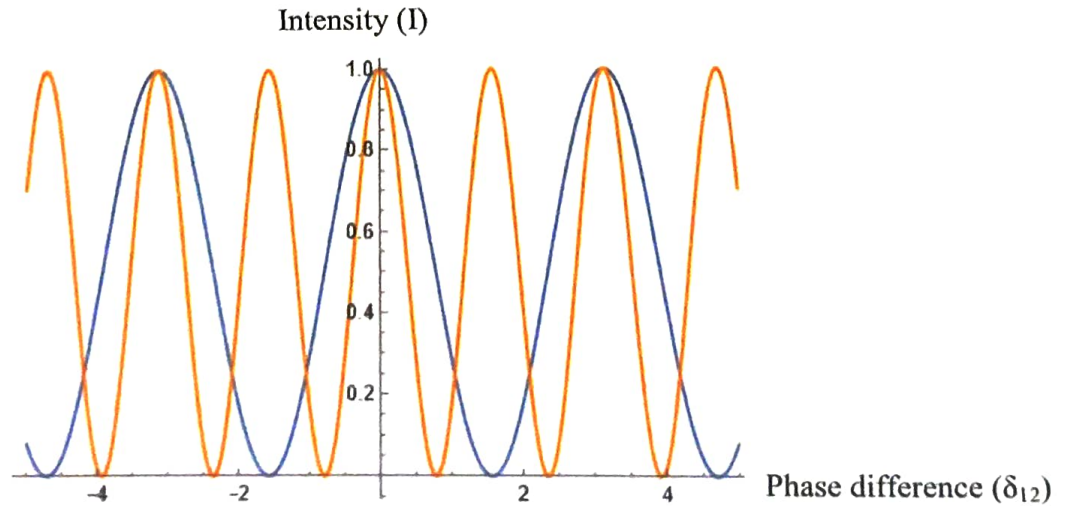


Figure-1: Blue color: Gaussian beam ($l = 0$); Orange color: LG beam ($l = 1$)

Figure-1 is the graphical representation of the relation between intensity and phase (azimuthal phase) difference for the Gaussian (LG) beam(s). It can be observed from figure-1 that between δ (-4 to 4) Gaussian beam ($l=0$) has three constructive and two destructive interference while for LG beam ($l=1$) we have observed five constructive and six destructive interference. If we use LG beam with $l=2$ or more then the number of constructive and destructive interference increases as the value of l increases. Thus we can say the by controlling the value of OAM we can manipulate the interference pattern.

CONTOUR PLOT

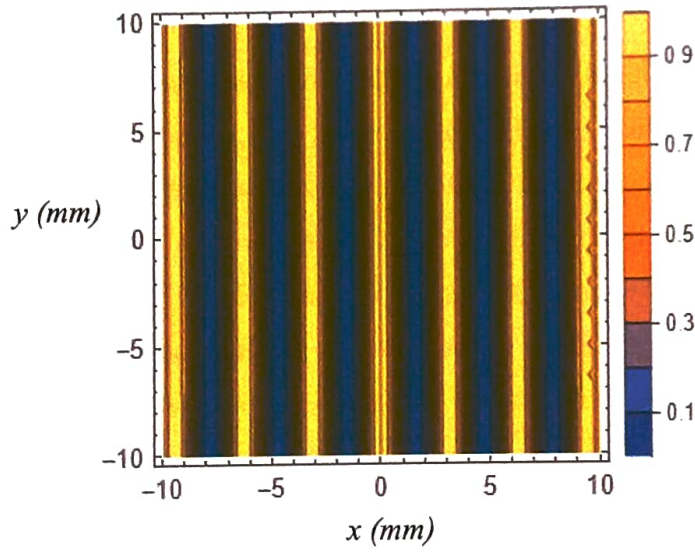


Figure 2: Gaussian beam ($l = 0$)

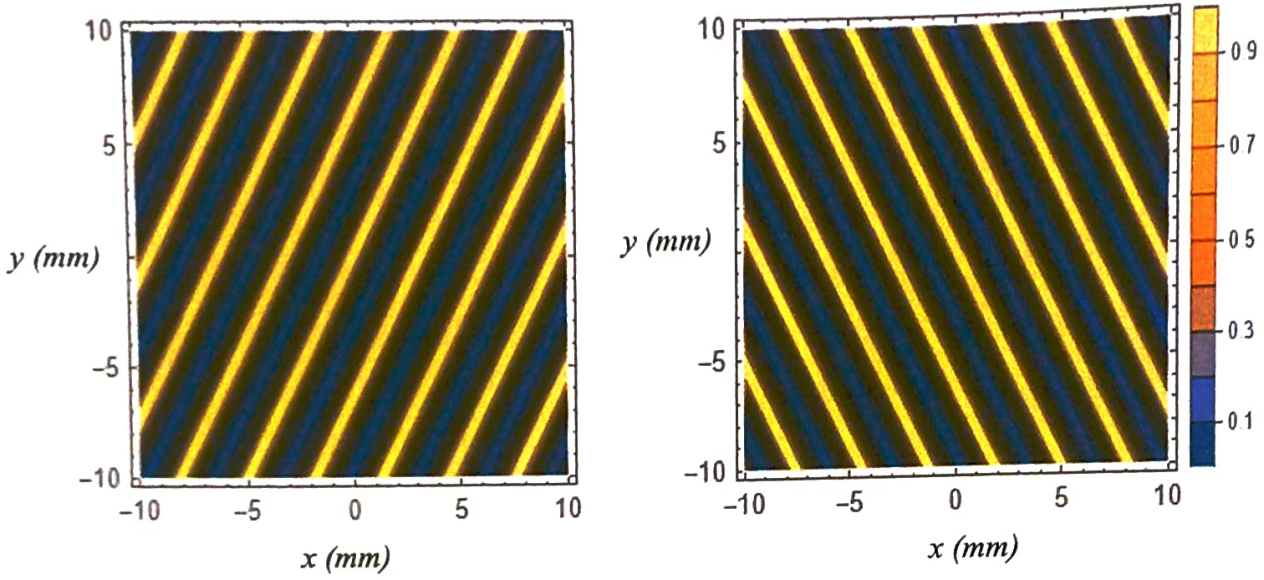
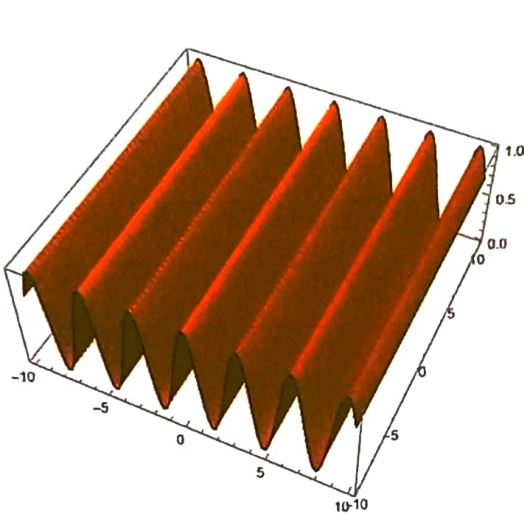


Figure 3: (a) LG beam ($l=1$)

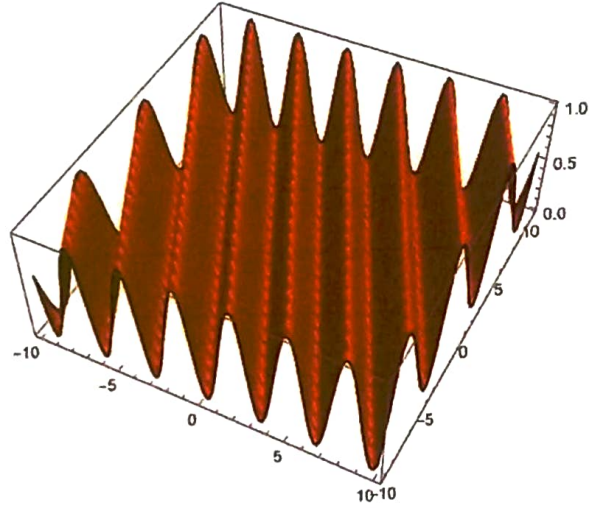
(b) LG beam ($l=-1$)

In order to understand the better insight view of the interference pattern formed by two Gaussian beam as well as two LG beams, we have picturized the two-dimensional profile of the interference pattern. For this purpose, we have depicted the intensity of the interference fringe against the spatial axes x, y . Figure 2 & 3 are the spatially dependent behavior profile of the interference fringe plot for a Gaussian wave ($l=0$) and LG beam with the OAM value of $l=\pm 1$. Comparing the contour plot for Gaussian with that of LG we can see the shift of the line for LG but it is straight line for Gaussian wave. It is found that the azimuthally dependence part of the LG beam gives additional phase shift of the interference pattern. Thus, the positions of both the dark and bright fringes are shifted in comparison to the result for Gaussian beam. It is also found that the outcomes for $l=+1$ give the same information in comparison to $l=-1$. The spatially dependent behaviour profiles of the interference pattern between two LG beams indicate the angular position of both the dark and bright fringes are possible to manipulate by the OAM number.

THREE DIMENSIONAL PLOT



GAUSSIAN BEAM ($l=0$)



LG BEAM ($l=1$)

CONCLUSION

In this project, we have studied the interference fringe patterns produced by two Gaussian beams as well as the two LG beams. The detail theoretical background of the inference phenomena has presented by considering the beams to be Gaussian or LG beam. Then we have compared both results. It is found that we can control the interference pattern formed by two LG beams with the help of controlling OAM number. The number of both bright and dark fringes within a certain interval can be manipulated through the OAM number. In conclusion, we may claim that the OAM number can be used as a control knob to manipulate the number of fringes as well as the positions. This is a major advantage when someone is working with LG beam in compared to a Gaussian beam. Moreover, we see the effect of the azimuthal angle of the LG beam on the interference fringes. This work has demonstrated that by using LG beam one can control both the position and number of the interference fringes.

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